Homework Assignment 1

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Rutgers University

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Problem 1:

(a) Let sample space be defined by

$$S = \{1, 2, 3, 45, 6\}$$

(b)

$$A \cap B = \{1, 3\}$$
$$(A \cup B)^c = \{4, 6\}$$

(c)

$$P(A \cap B) = \frac{|A \cap B|}{|S|} = \frac{1}{3}$$
$$P((A \cup B)^c) = \frac{|(A \cup B)^c|}{|S|} = \frac{1}{3}$$

Problem 2: Using the multinomial coefficient, we get

$$\binom{11}{2\ 4\ 4} = \frac{11!}{4! * 4! * 2!} = 34,650 \text{ different letter arrangements}$$

Problem 3:

(a)

$$\binom{8}{5} - \binom{6}{3} = 56 - 20 = 36 \text{ choices}$$

(b)

$$\binom{6}{5} + \binom{6}{3} = 6 + 20 = 26 \text{ choices}$$

Problem 4:

(a) Using stars and bars we get

$$\binom{13}{4} = 715 \text{ possible divisions}$$

(b) We subtract 5 from n since each school must receive one and there is 5 schools

$$\binom{8}{4} = 70 \text{ possible divisions}$$

Problem 5:

(a)

$$15^3 = 3375 \text{ different cones}$$

(b)

$$P(15, 3) = 2730 \text{ different cones}$$

(c)

$$\binom{17}{3} = 680 \text{ different cones}$$

(d)

$$\binom{15}{3} = 445 \text{ different cones}$$

Problem 6:

(a)

$$8! = 40320 \text{ different seating arrangements}$$

(b)

$$2! * 7! = 100800 \text{ different seating arrangements}$$

(c)

$$2! * 4! * 4! = 1152 \text{ different seating arrangements}$$

(d)

$$4! * 5! = 2880 \text{ different seating arrangements}$$

(e)

$$4! * (2!)^4 = 384 \text{ different seating arrangements}$$